

TUSHAR JAYENDRA MHATRE

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PROFESSIONAL SUMMARY

Data Scientist/ML Engineer with 3.5 years of experience in designing, deploying, and optimizing end-to-end machine learning and AI solutions. Skilled in MLOps, predictive modeling, and ETL pipelines, with a strong foundation in A/B testing, statistical analysis, and cloud-based ML deployment.

PROFESSIONAL EXPERIENCE

The Aether Loop

AI Engineering Intern

Remote

September 2025 - Present

- Building multi-Agent AI system with Crew AI to build a robust supplement recommendation system based on patient blood biomarkers.
- Containerized and deployed backend services on Google Cloud Run with Cloud Build and CI/CD, hosted LLM endpoint on VertexAI.
- Developed edge functions in javascript to make the API calls from supabase to GCP, integrating multiple backends seamlessly.

Optimal living Systems

Data Architect Intern

Remote

July 2025 - September 2025

- Developed and orchestrated ETL pipelines in Airflow that processed 1,000+ research PDFs into structured synthetic datasets through OCR(Docling) and open-source LLMs for LLM fine-tuning.
- Optimized synthetic data storage in GCP BigQuery with schema-structured collections, cutting LLM training data prep time by 40%.

University of Oklahoma

Graduate Research Assistant

Oklahoma, USA

February 2024 - May 2025

- Developed automated pipelines for transforming EEG and MRI data into BIDS-compliant structures in Python and MATLAB, cutting the manual conversion time by 40% and streamlining lab-wide data workflows.
- Maintained and updated large-scale neuroimaging datasets, ensuring seamless accessibility for a lab with 15 researchers.
- Applied clustering algorithms to identify Cortical Activity Patterns (CAPs) in brain EEG data across 10+ studies.
- Conducted statistical analysis (hypothesis testing, causal inference) on neuroimaging data to validate numerous research findings.
- Developed WSL-based Bash scripts to bridge Windows/MATLAB tools with Linux-optimized neuroimaging pipelines (FSL, AFNI), cutting preprocessing runtime by 40% through parallel job scheduling.
- Presented findings for research papers and presentations, translating complex findings for non-technical audiences.

Tata consultancy Services

Analyst

Pune, Maharashtra, India

February 2021 - July 2023

- Collaborated with cross-functional teams in Agile sprints to gather requirements, define project scope, and deliver data solutions aligned with business needs.
- Used MLflow to track model performance, hyperparameters, and artifacts across 20+ forecasting experiments, enabling reproducible model selection and reducing deployment cycle time by 30%.
- Designed and deployed ETL pipelines (airflow) to aggregate data from multiple sources (SAP, mainframes) for forecasting models.
- Diagnosed and resolved 30+ production incidents in live forecasting pipelines (Python, Airflow), implementing fixes that reduced error rates by 30% and improved system uptime significantly.
- Developed and containerized ML models (Docker) and exposed as microservices via Flask, reducing prediction latency from 2secs to 300ms while handling 1K+ requests per minute (RPM).
- Conducted rigorous A/B testing on model iterations, demonstrating 15% lower RMSE and 20% better bias than previous models.
- Implemented CI/CD pipelines (GitHub Actions) to automate testing and deployment of ML models

EDUCATION

University of Oklahoma

Master's, Data Science

August 2023 - May 2025

GPA: 3.97

- Courses: Data mining, Machine Learning Practice, Data Visualization, Database,data structures, Healthcare analytics, Statistics

PROJECTS

AI Research Paper Assistant - [Link to project](#)

- Developed a LLM-based multimodal research chatbot with Retrieval Augmented generation implemented with FAISS, Ollama, Langchain, and Python.
- Used Unstructured and docling for OCR and text extraction, created textual description and embeddings for figures and tables, and implemented hybrid retrieval for accurate retrieval.
- Built a UI in Streamlit for seamless interaction with the system and optimized the model through prompt engineering.

SKILLS

- Programming Languages:** Python, R, SQL, C/C++, MATLAB, Java
- Python Libraries:** NumPy, Pandas, matplotlib, Scikit-learn, Tensorflow, XGboost, Pytorch, Seaborn, imblearn, Flask
- Statistical Analysis:** Hypothesis Testing, Confidence Intervals, Outlier Detection, Causal Inference
- Database:** MySQL, Database Design, JDBC (Java Database Connectivity), Microsoft Azure, FAISS, PostgreSQL, GCP BigQuery
- Business Intelligence tools & Technologies:** Tableau, Microsoft Power BI, Excel/Numbers/Sheets, Git, Jupyter
- Natural Language Processing:** Tokenization, Named Entity Recognition (NER), TF-IDF, Word Embeddings (Word2Vec)
- Generative AI:** LangChain, Ollama, Langraph, Crew AI, Hugging Face, RAG, Unsloth AI